

Teaching a Translation Model a Language It Never Knew: Ekegusii for Kenyan Public Service Announcements

NLLB-200 supports 200 languages. **Ekegusii is not one of them.** Using **transfer learning**, we added `guz_Latn` to the model and fine-tuned it on **62,669 parallel sentence pairs**, so that Kenyan public service announcements can reach 2.7 million more speakers.

WHAT IT LEARNED FROM



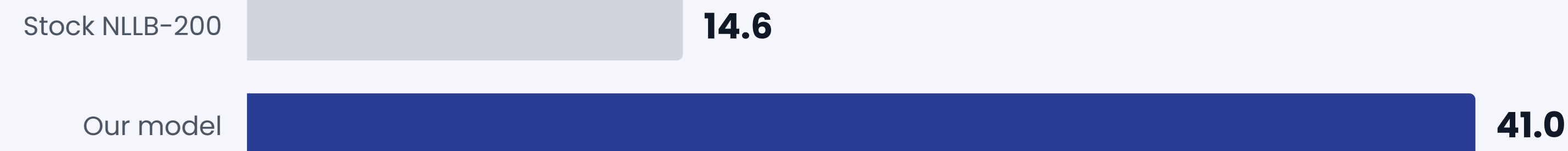
Ekegusii Bible	56,866	90.7%
Kenyan public service announcements	5,692	9.1%
Everyday sentences	111	0.2%

RESULT: chrF2++ ON HELD OUT PUBLIC SERVICE ANNOUNCEMENTS

40.97_{chrF2++}

English into Ekegusii

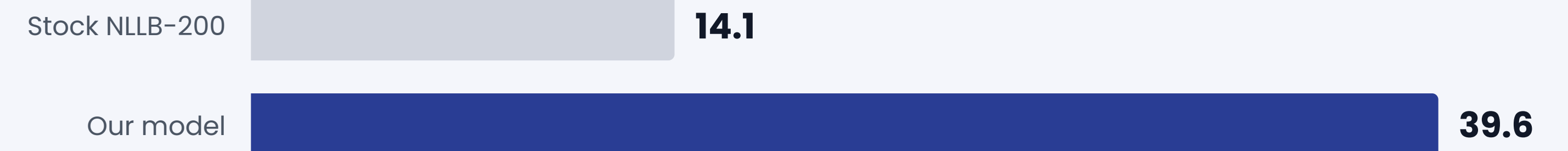
+26.4 over stock NLLB-200, a 181% relative gain



39.61_{chrF2++}

Kiswahili into Ekegusii

+25.5 over stock NLLB-200, a 180% relative gain



EXPERIMENTAL SETUP AND LIMITATIONS

How it was measured

chrF2++ on 570 English and 371 Kiswahili held out announcements. Stock NLLB-200 cannot produce Ekegusii and was asked for the nearest language it supports, so its bar is a floor rather than a baseline.

What was held back

2,993 scripture pairs, 200 everyday sentences and 944 public service announcements were never trained on. A further 27,575 pairs from a fourth source were discarded as duplicates of text already held.

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PROGRAMME

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CODE AND MODEL

github.com/SamAbr/public-service-announcement-MT